ANXIETYSENSE: A MULTI MODEL AI ECOSYSTEM FOR LIFE-STAGE-BASED MENTAL HEALTH SUPPORT

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PROBLEM AND IT'S IMPACT

- Anxiety levels are increasing across all life stages: children, young adults, pregnant women, and the elderly
- Existing assessments rely mainly on self-reported questionnaires
- Users may give incomplete, biased, or inaccurate responses
- Emotional distress is often hidden in facial expressions and voice patterns
- Traditional assessments are time-consuming, subjective, and clinic-dependent
- In Sri Lanka, mental health stigma and limited access worsen the problem

WHO WILL BE BENEFITED

- Children – difficulty expressing emotions accurately
- Young Adults – academic, career, and social pressure
- Pregnant Women – anxiety hidden in voice and emotional cues
- Elderly (60+) – struggle with complex digital tools

DESIGN EXCELLENCE

- Multimodal AI system combining:
  - Questionnaires
  - Facial emotion recognition
  - Voice emotion analysis
- Life-stage–specific anxiety models for better accuracy
- Real-time, non-intrusive emotion detection
- Simple, clean, and emotionally supportive user interfaces

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MULTIMODAL CHILD ANXIETY MANAGEMENT ANALYSIS

MULTIMODAL CHILD ANXIETY MANAGEMENT ANALYSIS

PROBLEM

- Rising levels of anxiety among children
- Current anxiety assessments rely mainly on self-reported questionnaires (SCAS)
- Children may give incomplete, inaccurate, or biased responses
- Single-modal systems (only questionnaire or only face) show limited accuracy
- Limited use of multimodal AI approaches combining questionnaire and facial emotion analysis

USER REQUIREMENTS

- The system must allow the child to answer the SCAS-19 questionnaire
- The system must request camera permission from the user
- The system must capture facial images/video for emotion analysis
- Facial emotions must be detected using a MobileNet-based CNN model
- Questionnaire and facial emotion results must be processed securely

SOLUTION

- Develop a multimodal AI-based child anxiety assessment system
- Use SCAS-19 questionnaire for initial anxiety prediction
- Apply facial emotion recognition using MobileNet CNN
- Combine questionnaire results with facial emotion outputs
- Improve overall anxiety prediction accuracy through model fusion

DESIGN EXCELLENCE

- Facial images captured through the camera are processed in real time, providing instant emotion-based anxiety insights
- The system detects child facial emotions using a MobileNet-based CNN, ensuring fast and efficient emotion recognition
- Questionnaire-based anxiety prediction and facial emotion outputs are combined to enhance accuracy
- The system maintains a clean, child-friendly, and minimal UI suitable for young users
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USER FEEDBACK

- Users found the system easy to understand and simple to use
- Facial emotion detection was perceived as fast and responsive
- Reduced cognitive effort compared to questionnaire-only assessment
- Users felt the combined approach gave more realistic and reliable anxiety results

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MULTIMODAL YOUNG ADULT ANXIETY MANAGEMENT ANALYSIS

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MULTIMODAL YOUNG ADULT ANXIETY MANAGEMENT ANALYSIS

PROBLEM

- Rising anxiety and stress among young adults
- Mental health assessments rely mainly on self-reported questionnaires
- Users may give inaccurate or biased answers
- Limited use of AI-based emotion analysis in existing systems

SOLUTION

- Develop an AI-based mental health assessment system
- Use Dass 21 questionnaire for anxiety level prediction using random forest algorithm
- we got the questioneer data from the councilor
- Use facial emotion recognition with a CNN model
- we got the face data from the existing dataset
- Emotion-driven adaptive user interface
- Provide real-time emotion analysis using webcam/camera
- Deliver results through a user-friendly mobile/web interface

USER REQUIREMENTS

- The system must request camera permission from the user
- User must be able to complete the DASS-21 questionnaire
- The system must capture facial images and detect emotions
- User must receive a final anxiety level assessment

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DESIGN EXCELLENCE

- Facial images captured through the camera are processed instantly, providing immediate emotion prediction to the user
- The system detects user emotions in real time using a custom-trained CNN model without relying on pretrained architectures.
- The UI dynamically changes colors based on the detected emotion to support emotional regulation.
- DASS-21 is used for structured mental health assessment, while emotion detection is used for emotional support through UI adaptation

USER FEEDBACK

- Found the system easy to understand and use
- Emotion detection was perceived as fast and responsive
- Reduced effort compared to answering questionnaires alone
- Felt the system gave more realistic results
- Positive feedback on clean and simple user interface

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PREGNANCY – SPECIFIC ANXIETY DETECTION WITH VOICE ANALYSIS

PREGNANCY – SPECIFIC ANXIETY DETECTION WITH VOICE ANALYSIS

PROBLEM

- Difficulty expressing emotions through questionnaires
- Emotional distress hidden in voice patterns
- Manual assessments are time-consuming and subjective
- No real-time, non-intrusive emotion detection

SOLUTION

- Analyze voice recordings using audio processing
- Detect emotions from voice input using LSTM model
- Reject empty or low-volume recordings
- Provide instant emotional feedback
- Combines voice-based emotion results with PASS questionnaire scores
- Produces a final anxiety prediction level with improved accuracy

USER REQUIREMENTS

- Easy voice recording
- Clear error messages for invalid inputs
- Fast and accurate emotion results
- Simple, non-medical output
- Secure and private voice data handling

DESIGN EXCELLENCE

- Uses MFCC feature extraction for accurate voice pattern analysis
- Integrated emotion classification model for reliable emotion prediction
- Frontend and backend validation to prevent invalid voice inputs
- Lightweight and scalable architecture suitable for real-time usage
- Modular design allows easy integration with other system modules
- Non-intrusive voice-based approach improves user comfort and acceptance

USER FEEDBACK

- Users found voice input easy and comfortable
- Error messages improved clarity for low or empty voice inputs
- Emotion results were understandable and helpful
- Overall experience felt non-judgmental and supportive

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ANXIETYSENSE: AI-POWERED, CULTURALLY-ADAPTED MENTAL HEALTH SUPPORT FOR SRI LANKA'S ELDERLY

OLD ADULTS - SPECIFIC ANXIETY DETECTION WITH PERSONALIZED EMOTIONALLY MUSIC THERAPY

PROBLEM

- Anxiety in adults aged 60+ is often underdiagnosed
- Elderly users struggle with complex digital tools
- Traditional assessments are time-consuming & clinic-dependent
- Lack of culturally relevant emotional support in Sri Lanka

SOLUTION

- AI-assisted anxiety detection designed specifically for 60+ adults
- Combines:

1. Geriatric Anxiety Scale (GAS)
2. Facial emotion recognition
3. Culturally familiar music therapy

- Elder-friendly, simple, and non-intrusive system

USER REQUIREMENTS

- Large text and clear buttons
- Sinhala & English language support
- Minimal steps to complete assessment
- Privacy-safe and non-judgmental feedback

DESIGN EXCELLENCE

- GAS integrated with real-time emotion recognition
- CNN models optimized for elderly facial features
- On-device processing for privacy protection
- Simple, calming UI designed for seniors

USER FEEDBACK

- Elderly users found the app easy to follow
- Music therapy felt familiar and comforting
- Anxiety results were easy to understand
- Interface felt supportive, not clinical
- Users preferred this over hospital visits

THANK YOU

Feel free to share, support and help each other!